Lösung Übung 1

18 October 2025 13:34

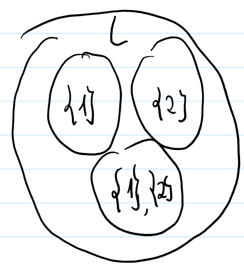
Aufgabe 1 $M = \{1,2,3\}$

a) (i) $1 \in M$ ✓ richtig

(ii) $\{1\} \in M$ ✗ falsch $\{1\} \subseteq M$

(iii) $\{1\} \subseteq M$ ✓ richtig

b) $L = \{\emptyset, \{1\}, \{2\}, \{1,2\}\}$ menge in menge



(i) $2 \in L$ ✗ $2 \in \{2\}$ or $\{2\} \in L$

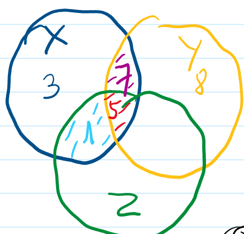
Element in Menge Menge Menge in Menge

(ii) $\{2\} \in L$ ✓ richtig

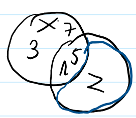
(iii) $\{2\} \subseteq L$ ✗ falsch, $\{2\}$ ist nicht Teilmenge von L

(iii) $\{\{2\}\} \subseteq L$ ✓ richtig

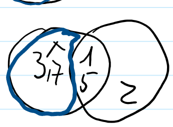
Aufgabe 2 $X = \{1,3,5,7\}$, $Y = \{5,7,8\}$, $Z = \{1,5\}$



(i) $Z \setminus X = \{\} = \emptyset$

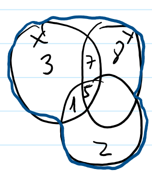


(iii) $X \setminus Z = \{3,7\}$



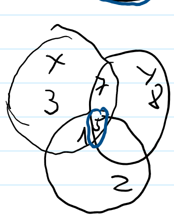
(iv) $X \cup Y \cup Z$

$= \{1,3,5,7,8\}$



(iiii) $X \cap Y \cap Z$

$= \{5\}$



(V) $X \times Z$ X Z

Binäre Relation $1 \longrightarrow 1$ $\Rightarrow (1,1), (1,5)$

(v) $X \times Z$

Binäre Relation

$$\begin{array}{ccc} x & & z \\ 1 & \longrightarrow & 1 \\ & \searrow & 5 \end{array} \Rightarrow (1,1), (1,5)$$

$$\begin{array}{ccc} 3 & \nearrow & 1 \\ & \searrow & 5 \end{array} \Rightarrow (3,1), (3,5)$$

$$\begin{array}{ccc} 5 & \nearrow & 1 \\ & \searrow & 5 \end{array} \Rightarrow (5,1), (5,5)$$

$$\begin{array}{ccc} 7 & \nearrow & 1 \\ & \searrow & 5 \end{array} \Rightarrow (7,1), (7,5)$$

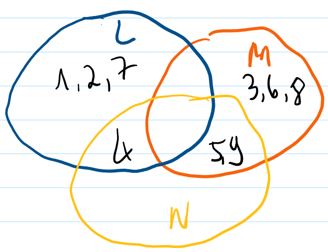
$$\Rightarrow X \times Z = \{(1,1), (1,5), (3,1), (3,5), (5,1), (5,5), (7,1), (7,5)\}$$

b) $G = \{1,2,3,4,5,6,7,8,9\}$

$L = \{1,2,4,7\}$

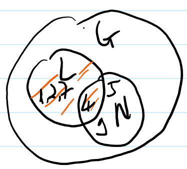
$M = \{3,5,6,8,9\}$

$N = \{4,5,9\}$



$G = L \cup M \cup N$

(i) $\overline{L} \cap N$



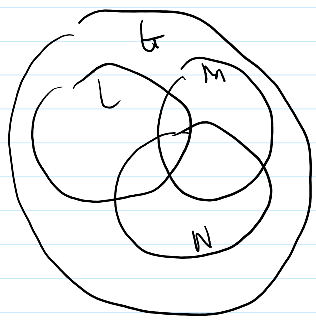
$= G \setminus L \cap N$

$= \{5, 9\}$

(ii)

$(L \cap \overline{M}) \cup (N \cap \overline{N})$

$= (L \cap (G \setminus M) \cup (N \cap \overline{N})$



- $G \setminus M = \{1,2,4,7\}$

  $L \cap (G \setminus M) = \{1,2,4,7\} \cup \{1,2,4,7\}$

  $= \{1,2,4,7\} = L$

- $N \cap \overline{N} = \{\} = \emptyset$

$\Rightarrow \quad L \cup \emptyset = \underline{\underline{L}}$

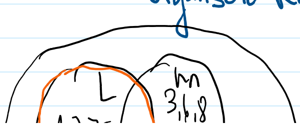
(iii) $L \cap \overline{N \cap \overline{M}}$

Morgansche Regeln $\overline{A \cup B} = \overline{A} \cap \overline{B}$

$\overline{A \cap B} = \overline{A} \cup \overline{B}$

$= L \cap \overline{N} \cup \overline{\overline{M}}$

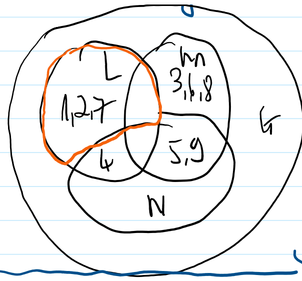
$L \cap (\overline{N} \cup M)$



$= L \cap \overline{N} \cup \overline{\overline{M}}$ $\qquad \overline{A \cap B} = \overline{A} \cup \overline{B}$

$= L \cap (\overline{N} \cup M)$

$= L \cap [(G \setminus N) \cup M]$



$\Downarrow$

$= L \cap (G \setminus N)$ → Weil M ist ein Teilmenge von G

$\uparrow \{1,2,3,6,7,8\}$

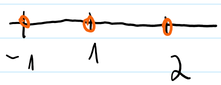
$= \{1,2,4,7\} \cap \{1,2,3,6,7,8\}$

$= \{1,2,7\}$

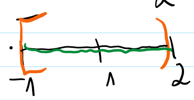
c) $M_1 = \mathbb{Z} \qquad M_2 = \mathbb{N}$

$M_3 = \{-1,1,2\} \quad M_4 = [-1,2)$

(i) M_3



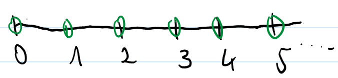
M_4



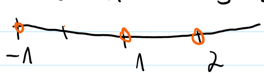
$\Rightarrow M_4 \cup M_3 = \{x \in \mathbb{R} : -1 \leq x < 2\}$

(ii) $M_4 \cap (M_2 \setminus M_3)$

M_2:

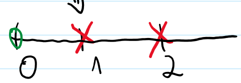


M_3:



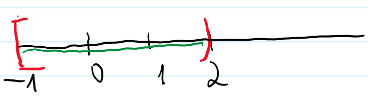
$\Downarrow$

$M_2 \setminus M_3$:



$\Rightarrow \{0\}$

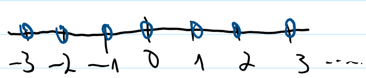
M_4



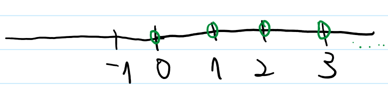
$M_4 \cap (M_2 \setminus M_3) = \{0\}$

(iii) $M_3 \setminus (M_1 \setminus M_2)$

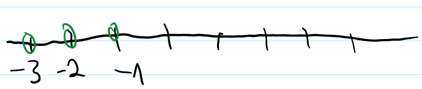
M_1



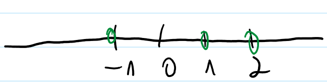
M_2

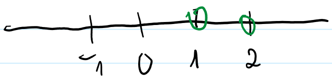


$M_1 \setminus M_2$



M_3



$M_3 \setminus (M_1 \setminus M_2)$  $\{1,2\}$

d) $A = \{1,2,3\}$

$A_0 = \{\}$

$A_1 = \{1\}$ $A_2 = \{2\}$ $A_3 = \{3\}$

$A_4 = \{1,2\}$ $A_5 = \{2,3\}$ $A_5 = \{1,3\}$

$A_6 = \{1,2,3\}$

Aufgabe 3

a) (i) ohne Rest durch 7 teilbar: $7n + 0 = 7n$

Rest ↑ 0

(ii) durch 5 Rest 3 $\Rightarrow 5n+3$

durch 5 ↑ $5n$

(iii) durch 2 und durch 3 ohne Rest

↑ $2n$ ↑ $3n$ ↑ 0

$2n \times 3n = 6n^2$ Optimation: $\underline{\underline{6n}}$

b)

x	$3n-2$	$3n+2$	2^n	2^{2n}
0	-2	2	1	1
1	1	5	2	4
2	4	7	4	16
3	7	11	8	64
4	10	14	16	256

(i) $\{3n-2 : n \in \mathbb{N}\} = \{-2, 1, 4, 7, 10 \ldots\}$

(ii) $\{3n+2 : n \in \mathbb{N}\} = \{2, 5, 7, 11, 14 \ldots\}$

(iii) $\{2^n : n \in \mathbb{N}\} = \{1, 2, 4, 8, 16 \ldots\}$

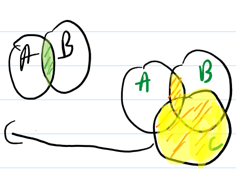
(iv) $\{2^{2n} : n \in \mathbb{N}\} = \{1, 4, 16, 64, 256\}$

Aufgabe 4

a)(i) $(A \cap B) \cup C$

$A \cap B$

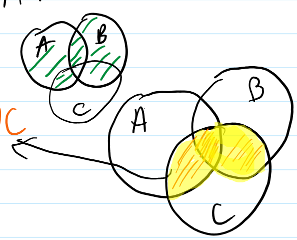
$(A \cap B) \cup C$



$(A \cup B) \cap C$

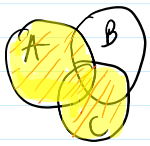
$A \cup B$

$(A \cup B) \cap C$

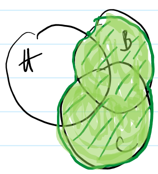


(ii) $(A \cup C) \cap (B \cup C)$ Distributivgesetze

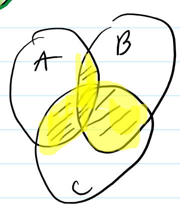
$A \cup C$



$B \cup C$

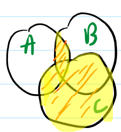


(iii) $(A \cup C) \cup (B \cup C)$

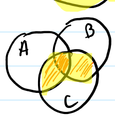


(iv) Distributivgesetze: $(A \cap B) \cup C = (A \cup C) \cap (B \cup C)$

(b) $A \cup (B \cap C)$

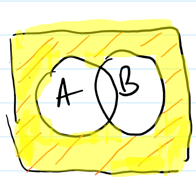


$(A \cup B) \cap C$

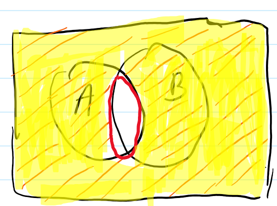


Es fällt auf, wenn die Klammern nicht weglassen kann

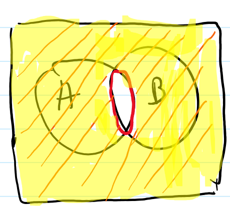
c) $\overline{A \cup B}$



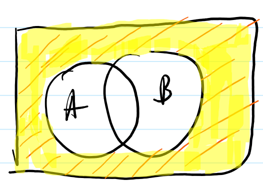
$\bar{A} \cup \bar{B}$



$\overline{A \cap B}$



$\bar{A} \cap \bar{B}$

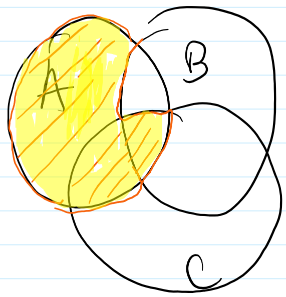


$\Rightarrow \overline{A \cup B} = \bar{A} \cap \bar{B}$

$\overline{A \cap B} = \bar{A} \cup \bar{B}$

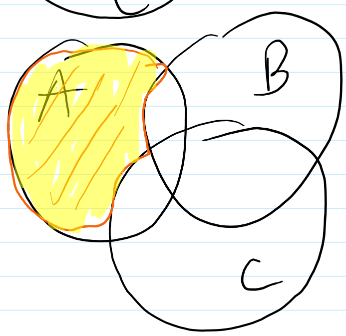
Aufgabe 5

(i) $A \setminus (B \setminus C)$



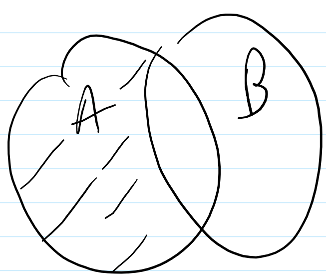
$\Rightarrow$ nicht erfüllt

$(A \setminus B) \setminus C$

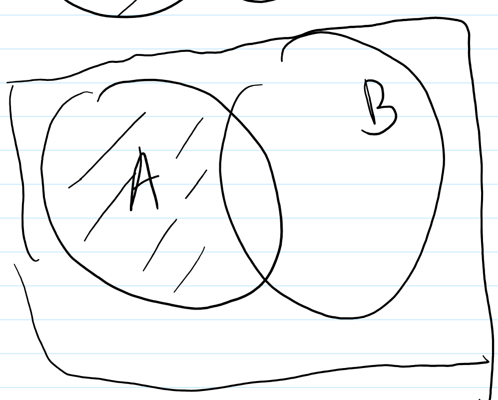


(ii)

$A \setminus B$

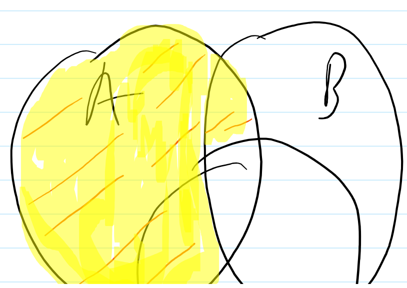


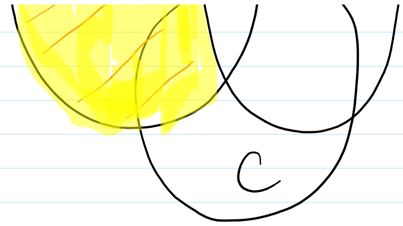
$A \cap \overline{B}$



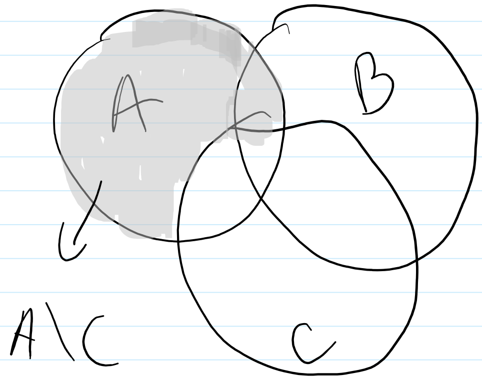
erfüllt

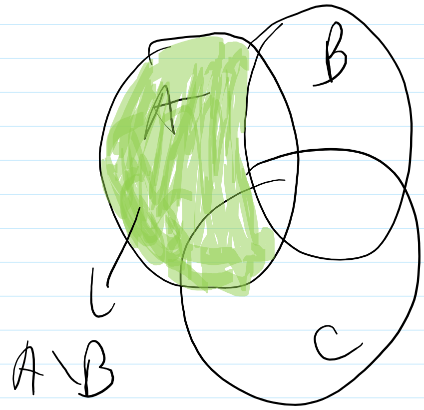
(iii) $A \setminus (B \cap C)$



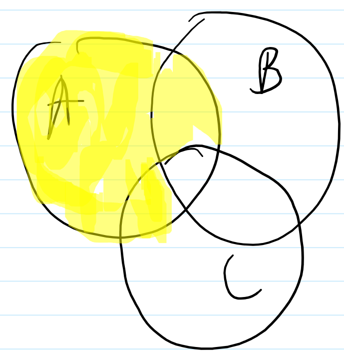


$(A\setminus B)\cup(A\setminus C)$





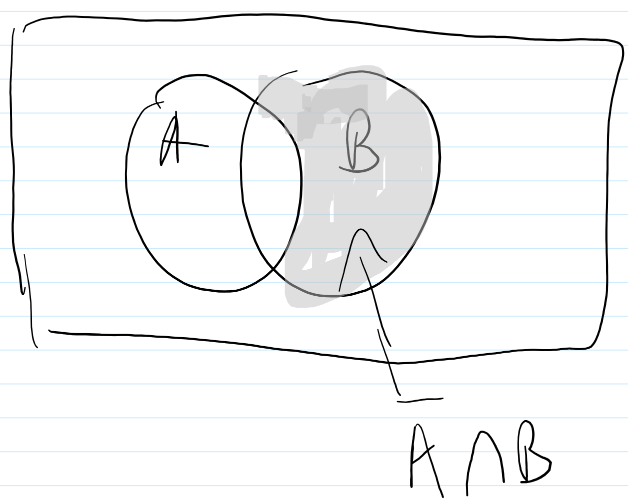
$(A\setminus B)\cup(A\setminus C)$



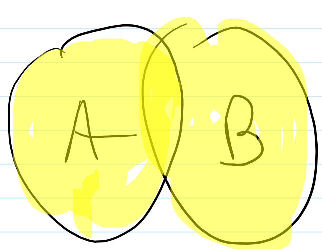
$\Rightarrow$ erfüllt

(iv)

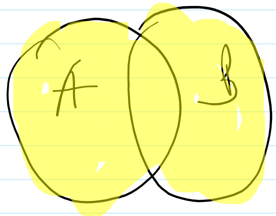
$A\cup(\bar{A}\cap B)$



$\Rightarrow A\cup(\bar{A}\cap B) \Rightarrow$



$A \cup B$



$\Rightarrow$ erfüllt